

Solution Requirements

Date	12 July 2026
Team ID	
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	No login is required in the current prototype; each student uses the app as an open, single-session Streamlit tool on their own device.	Low
2	Authorization levels	No differentiated user roles exist yet; every user has the same access to submit text and view their own analytics dashboard.	Low
3	External interfaces	Integrates with the Google Gemini API to generate natural, personalized guidance text based on the detected emotion and confidence score.	High
4	Transactions processing	Each student submission is processed as a single request: text input, emotion prediction, guidance generation, and CSV log write, completed within one Streamlit interaction cycle.	High
5	Reporting	Generates a live analytics dashboard with a bar chart of emotion-state distribution and a box plot comparing BiLSTM vs. BERT confidence, rendered via Plotly.	Medium
6	Business rules, etc.	Only the five defined emotion categories (Confused, Frustrated, Curious, Confident, Bored) drive response selection; any unmatched case falls back to a generic encouragement message.	High
7	Compliance to laws or regulations	No student personal identifiers are collected; only free-text input and predicted emotion/confidence are logged, limiting personal-data exposure.	Medium
8	Other	Supports side-by-side comparison of BiLSTM and BERT predictions for the same input via a <code>compare_models()</code> function.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Emotion prediction and guidance generation should feel conversational within a single Streamlit interaction.	Response returned in under 3 seconds per submission.
2	Scalability	The system should support a full classroom of students using the tool concurrently without slowdown.	Supports at least 50 concurrent Streamlit sessions in a classroom pilot.
3	Security & Data Privacy	Student text should not be exposed beyond the Gemini API call itself, and the API key must never enter source control.	GEMINI_API_KEY stored only in .env, excluded via .gitignore.
4	Reliability & Availability	If the Gemini API is unavailable, the platform should still return a helpful response rather than failing outright.	100% of requests return a response, falling back to the mock handler on API failure.
5	Usability & Accessibility	The interface should be usable by a student with no training, with clearly labeled emotion and guidance sections.	A first-time user can submit input and read guidance within 30 seconds, with no instructions needed.
6	Other	Maintainability: the four modules (emotion detection, AI response, logging, analytics) should remain independently testable and swappable (e.g., mock vs. real Gemini handler).	Each module can be unit-tested in isolation using mock dependencies.